

The Enterprise LLMOps Handbook

A production-ready, vendor-neutral reference companion and cookbook for AI architects and AI developers. **Content baseline: June 2026.**

This handbook is an **implementation manual**. It defines every concept you need to *learn, understand, and ship* enterprise-grade Large Language Model (LLM) systems, then backs each concept with copy-paste-ready configuration, manifests, pipelines, and checklists. It is deliberately **project-agnostic** — nothing here assumes a specific product, domain, or cloud. Use it as the spine for any new LLM initiative.

Who this is for

Reader	What you get
AI / Platform Architect	Reference architectures, decision trees, control mappings (OWASP, NIST AI RMF, EU AI Act), and progressive-delivery blueprints.
AI / ML Developer	Runnable examples: prompt registries, eval harnesses, guardrails, CI/CD, Helm, Argo Rollouts, Flagger, OpenTelemetry.
SRE / Platform Engineer	Runbooks, rollback decision trees, canary gates, drift & quality monitoring.
Security / GRC	Threat models, OWASP LLM Top 10 mappings, governance and compliance readiness.
Engineering Manager	Production-readiness checklists and a portable project-kickoff prompt to standardize delivery.

How to use this handbook

- Learning path** — read Part I → Part V in order to build vocabulary and controls.
- Cookbook path** — jump to the chapter for the capability you are building (e.g. `03-ragops.md`) and lift the examples.
- Delivery path** — start every new project by pasting `20-project-kickoff-prompt.md` into your AI coding agent (Codex, Claude Code, GitHub Copilot), scaffold the repo from `21-reference-repository-blueprint.md` , follow its phased sequence, and work the [production-readiness checklist](#) before go-live.

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Part XII — Reusable Asset

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Part XIII — Build It End-to-End

- [21-reference-repository-blueprint.md](#) — Complete reference repo tree, file-to-chapter matrix, and the phased Day-0→production implementation sequence

Conventions used throughout

- **Note** — helpful context.
- **Practice** — a recommended, defensible default.
- **Warning** — a failure mode or footgun to avoid.
- Code blocks are labeled with their language and are intended to be copy-paste-ready. Replace `{{PLACEHOLDERS}}` with your values.
- Diagrams use [Mermaid](#); math uses KaTeX (`$...$`).
- Every chapter ends with a **checklist** and a **References** section that links into [19-sources-and-references.md](#).

Authority basis

This handbook synthesizes public, authoritative sources — including the OWASP GenAI/LLM projects, the NIST AI Risk Management Framework and its Generative AI Profile, the EU Artificial Intelligence Act, the OpenTelemetry Generative-AI semantic conventions, CNCF projects (Kubernetes, Argo Rollouts, Flagger, Helm), the SLSA supply-chain framework, and recognized practitioner literature. Full citations are consolidated in [19-sources-and-references.md](#).

Warning — verify before you ship. Standards evolve. Treat version-specific details (API versions, regulatory deadlines, CVE guidance) as *as-of June 2026* and re-check against the primary source before a production decision.